Reversed achievement score in primary school for the oldest child

Current 2022/2023 enrollment defines the primary school sample. One oldest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.922	0.036 [0.069] N=1,935	0.054 [0.071] N=1,788	0.081 [0.076] N=1,612	0.095 [0.083] N=1,405	0.057 [0.092] N=1,178
Social studies and life sciences	3.937	0.053 [0.056] N=1,822	0.064 [0.057] N=1,685	0.076 [0.063] N=1,519	0.092 [0.068] N=1,327	0.044 [0.077] N=1,114
Mathematics	3.860	0.070 [0.075] N=1,912	0.108 [0.079] N=1,769	0.135 [0.084] N=1,594	**0.172* [0.091] N=1,390**	0.145 [0.100] N=1,170
Science	3.864	0.022 [0.082] N=1,383	0.020 [0.086] N=1,273	0.064 [0.092] N=1,139	0.057 [0.100] N=995	0.015 [0.114] N=838
Foreign language	3.665	0.080 [0.080] N=1,551	0.125 [0.083] N=1,434	0.116 [0.090] N=1,295	0.106 [0.099] N=1,133	0.089 [0.113] N=952
Culture arts and sports	3.997	**0.122* [0.066] N=1,616**	**0.127* [0.070] N=1,492**	**0.122* [0.073] N=1,350**	0.109 [0.082] N=1,177	0.083 [0.092] N=991
Vocational and technological	3.846	0.108 [0.143] N=615	0.113 [0.152] N=561	0.137 [0.167] N=513	0.102 [0.176] N=446	0.025 [0.200] N=370

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.029 [0.088] N=1,629	-0.006 [0.095] N=1,482	0.055 [0.109] N=1,306	0.077 [0.126] N=1,099	-0.066 [0.142] N=872
Social studies and life sciences	0.040 [0.080] N=1,529	0.049 [0.084] N=1,392	0.082 [0.096] N=1,226	0.123 [0.108] N=1,034	-0.018 [0.114] N=821
Mathematics	0.053 [0.090] N=1,607	0.109 [0.101] N=1,464	0.164 [0.113] N=1,289	**0.244* [0.130] N=1,085**	0.193 [0.159] N=865
Science	-0.035 [0.094] N=1,167	-0.053 [0.104] N=1,057	0.021 [0.115] N=923	-0.009 [0.132] N=779	-0.133 [0.160] N=622
Foreign language	0.037 [0.103] N=1,306	0.114 [0.110] N=1,189	0.093 [0.125] N=1,050	0.092 [0.141] N=888	0.028 [0.168] N=707
Culture arts and sports	0.131 [0.098] N=1,361	0.127 [0.110] N=1,237	0.139 [0.124] N=1,095	0.140 [0.155] N=922	0.076 [0.192] N=736
Vocational and technological	0.182 [0.185] N=532	0.197 [0.203] N=478	0.249 [0.245] N=430	0.244 [0.258] N=363	0.182 [0.326] N=287

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.922	0.045 [0.069] N=1,935	0.066 [0.071] N=1,788	0.094 [0.076] N=1,612	0.113 [0.082] N=1,405	0.077 [0.091] N=1,178
Social studies and life sciences	3.937	0.060 [0.056] N=1,822	0.072 [0.057] N=1,685	0.085 [0.063] N=1,519	0.105 [0.068] N=1,327	0.059 [0.077] N=1,114
Mathematics	3.860	0.068 [0.075] N=1,912	0.106 [0.079] N=1,769	0.134 [0.083] N=1,594	**0.169* [0.091] N=1,390**	0.140 [0.100] N=1,170
Science	3.864	0.023 [0.081] N=1,383	0.022 [0.085] N=1,273	0.069 [0.092] N=1,139	0.067 [0.099] N=995	0.026 [0.113] N=838
Foreign language	3.665	0.088 [0.079] N=1,551	0.133 [0.082] N=1,434	0.127 [0.090] N=1,295	0.119 [0.098] N=1,133	0.109 [0.111] N=952
Culture arts and sports	3.997	**0.127* [0.066] N=1,616**	**0.132* [0.069] N=1,492**	**0.128* [0.073] N=1,350**	0.119 [0.081] N=1,177	0.096 [0.091] N=991
Vocational and technological	3.846	0.103 [0.144] N=615	0.111 [0.153] N=561	0.130 [0.168] N=513	0.086 [0.178] N=446	0.022 [0.200] N=370

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.041 [0.087] N=1,629	-0.016 [0.093] N=1,482	0.041 [0.107] N=1,306	0.062 [0.123] N=1,099	-0.077 [0.137] N=872
Social studies and life sciences	0.031 [0.079] N=1,529	0.039 [0.083] N=1,392	0.070 [0.095] N=1,226	0.111 [0.107] N=1,034	-0.022 [0.112] N=821
Mathematics	0.055 [0.090] N=1,607	0.110 [0.101] N=1,464	0.164 [0.114] N=1,289	**0.246* [0.131] N=1,085**	0.194 [0.160] N=865
Science	-0.046 [0.094] N=1,167	-0.065 [0.103] N=1,057	0.005 [0.114] N=923	-0.017 [0.130] N=779	-0.135 [0.158] N=622
Foreign language	0.024 [0.104] N=1,306	0.100 [0.111] N=1,189	0.076 [0.126] N=1,050	0.074 [0.142] N=888	0.018 [0.170] N=707
Culture arts and sports	0.126 [0.098] N=1,361	0.123 [0.110] N=1,237	0.134 [0.124] N=1,095	0.132 [0.154] N=922	0.067 [0.192] N=736
Vocational and technological	0.166 [0.191] N=532	0.185 [0.210] N=478	0.218 [0.259] N=430	0.192 [0.281] N=363	0.158 [0.346] N=287

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.922	0.040 [0.068] N=1,935	0.057 [0.070] N=1,788	0.082 [0.074] N=1,612	0.091 [0.080] N=1,405	0.048 [0.088] N=1,178
Social studies and life sciences	3.937	0.057 [0.055] N=1,822	0.063 [0.056] N=1,685	0.070 [0.061] N=1,519	0.079 [0.067] N=1,327	0.021 [0.077] N=1,114
Mathematics	3.860	0.058 [0.074] N=1,912	0.090 [0.078] N=1,769	0.112 [0.082] N=1,594	0.135 [0.089] N=1,390	0.101 [0.099] N=1,170
Science	3.864	0.005 [0.081] N=1,383	0.000 [0.086] N=1,273	0.033 [0.093] N=1,139	0.022 [0.101] N=995	-0.015 [0.114] N=838
Foreign language	3.665	0.087 [0.080] N=1,551	0.126 [0.082] N=1,434	0.120 [0.090] N=1,295	0.103 [0.098] N=1,133	0.091 [0.113] N=952
Culture arts and sports	3.997	**0.119* [0.065] N=1,616**	**0.122* [0.068] N=1,492**	0.109 [0.072] N=1,350	0.089 [0.080] N=1,177	0.067 [0.090] N=991
Vocational and technological	3.846	0.069 [0.147] N=615	0.055 [0.157] N=561	0.041 [0.174] N=513	0.002 [0.185] N=446	-0.049 [0.202] N=370

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.051 [0.088] N=1,629	-0.036 [0.094] N=1,482	0.014 [0.107] N=1,306	0.007 [0.119] N=1,099	-0.160 [0.130] N=872
Social studies and life sciences	0.027 [0.080] N=1,529	0.025 [0.083] N=1,392	0.043 [0.095] N=1,226	0.058 [0.103] N=1,034	-0.113 [0.105] N=821
Mathematics	0.036 [0.089] N=1,607	0.078 [0.101] N=1,464	0.118 [0.113] N=1,289	0.165 [0.124] N=1,085	0.087 [0.155] N=865
Science	-0.062 [0.094] N=1,167	-0.089 [0.102] N=1,057	-0.046 [0.114] N=923	-0.096 [0.132] N=779	-0.211 [0.155] N=622
Foreign language	0.026 [0.107] N=1,306	0.090 [0.113] N=1,189	0.061 [0.129] N=1,050	0.032 [0.142] N=888	-0.023 [0.182] N=707
Culture arts and sports	0.109 [0.095] N=1,361	0.096 [0.104] N=1,237	0.086 [0.117] N=1,095	0.039 [0.144] N=922	-0.046 [0.176] N=736
Vocational and technological	0.157 [0.196] N=532	0.149 [0.217] N=478	0.131 [0.273] N=430	0.141 [0.305] N=363	0.159 [0.364] N=287

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.922	0.033 [0.067] N=1,935	0.044 [0.069] N=1,788	0.061 [0.072] N=1,612	0.066 [0.078] N=1,405	0.014 [0.085] N=1,178
Social studies and life sciences	3.937	0.047 [0.055] N=1,822	0.050 [0.057] N=1,685	0.049 [0.062] N=1,519	0.055 [0.067] N=1,327	-0.010 [0.078] N=1,114
Mathematics	3.860	0.053 [0.073] N=1,912	0.082 [0.077] N=1,769	0.097 [0.082] N=1,594	0.123 [0.089] N=1,390	0.082 [0.098] N=1,170
Science	3.864	0.000 [0.080] N=1,383	-0.010 [0.085] N=1,273	0.014 [0.092] N=1,139	0.001 [0.100] N=995	-0.052 [0.113] N=838
Foreign language	3.665	0.080 [0.079] N=1,551	0.118 [0.082] N=1,434	0.104 [0.089] N=1,295	0.085 [0.097] N=1,133	0.064 [0.111] N=952
Culture arts and sports	3.997	**0.108* [0.064] N=1,616**	0.106 [0.068] N=1,492	0.084 [0.071] N=1,350	0.064 [0.079] N=1,177	0.029 [0.088] N=991
Vocational and technological	3.846	0.096 [0.144] N=615	0.071 [0.155] N=561	0.060 [0.173] N=513	0.017 [0.185] N=446	-0.055 [0.204] N=370

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.057 [0.085] N=1,629	-0.052 [0.093] N=1,482	-0.013 [0.106] N=1,306	-0.031 [0.118] N=1,099	**-0.221* [0.131] N=872**
Social studies and life sciences	0.022 [0.078] N=1,529	0.010 [0.083] N=1,392	0.015 [0.095] N=1,226	0.024 [0.103] N=1,034	-0.164 [0.105] N=821
Mathematics	0.036 [0.088] N=1,607	0.067 [0.101] N=1,464	0.097 [0.112] N=1,289	0.153 [0.125] N=1,085	0.064 [0.155] N=865
Science	-0.053 [0.093] N=1,167	-0.093 [0.104] N=1,057	-0.059 [0.116] N=923	-0.109 [0.135] N=779	-0.249 [0.155] N=622
Foreign language	0.027 [0.104] N=1,306	0.087 [0.111] N=1,189	0.046 [0.125] N=1,050	0.010 [0.138] N=888	-0.070 [0.176] N=707
Culture arts and sports	0.108 [0.097] N=1,361	0.086 [0.107] N=1,237	0.064 [0.120] N=1,095	0.026 [0.148] N=922	-0.093 [0.181] N=736
Vocational and technological	0.224 [0.187] N=532	0.204 [0.209] N=478	0.213 [0.260] N=430	0.224 [0.291] N=363	0.214 [0.348] N=287

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.910	0.010 [0.069] N=1,822	0.019 [0.072] N=1,683	0.047 [0.077] N=1,515	0.067 [0.083] N=1,322	0.014 [0.092] N=1,107
Social studies and life sciences	3.926	0.048 [0.057] N=1,716	0.047 [0.058] N=1,585	0.056 [0.063] N=1,426	0.075 [0.066] N=1,247	0.027 [0.076] N=1,046
Mathematics	3.848	0.052 [0.079] N=1,802	0.079 [0.084] N=1,665	0.097 [0.089] N=1,498	0.131 [0.098] N=1,308	0.106 [0.109] N=1,100
Science	3.847	0.010 [0.083] N=1,294	-0.002 [0.088] N=1,188	0.044 [0.095] N=1,060	0.042 [0.103] N=927	0.013 [0.116] N=778
Foreign language	3.644	0.052 [0.088] N=1,460	0.101 [0.091] N=1,348	0.094 [0.099] N=1,214	0.094 [0.108] N=1,064	0.082 [0.122] N=892
Culture arts and sports	3.997	**0.138** [0.068] N=1,548**	**0.140* [0.071] N=1,428**	**0.133* [0.074] N=1,292**	0.133 [0.081] N=1,129	0.118 [0.092] N=948
Vocational and technological	3.840	0.115 [0.144] N=593	0.108 [0.152] N=542	0.132 [0.169] N=494	0.066 [0.185] N=434	0.025 [0.208] N=359

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.045 [0.089] N=1,539	-0.033 [0.095] N=1,400	0.022 [0.109] N=1,232	0.039 [0.128] N=1,039	-0.108 [0.145] N=824
Social studies and life sciences	0.035 [0.083] N=1,445	0.028 [0.086] N=1,314	0.050 [0.099] N=1,155	0.085 [0.112] N=976	-0.033 [0.114] N=775
Mathematics	0.056 [0.095] N=1,520	0.100 [0.106] N=1,383	0.138 [0.118] N=1,216	0.220 [0.138] N=1,026	0.191 [0.166] N=818
Science	-0.060 [0.099] N=1,098	-0.096 [0.108] N=992	-0.038 [0.121] N=864	-0.072 [0.139] N=731	-0.148 [0.165] N=582
Foreign language	0.002 [0.114] N=1,237	0.082 [0.122] N=1,125	0.048 [0.140] N=991	0.053 [0.158] N=841	0.014 [0.188] N=669
Culture arts and sports	0.120 [0.103] N=1,308	0.109 [0.115] N=1,188	0.107 [0.129] N=1,052	0.109 [0.159] N=889	0.058 [0.200] N=708
Vocational and technological	0.189 [0.179] N=515	0.175 [0.195] N=464	0.215 [0.242] N=416	0.189 [0.276] N=356	0.196 [0.331] N=281

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.910	0.003 [0.071] N=1,822	0.019 [0.073] N=1,683	0.047 [0.078] N=1,515	0.066 [0.084] N=1,322	0.013 [0.093] N=1,107
Social studies and life sciences	3.926	0.045 [0.058] N=1,716	0.049 [0.059] N=1,585	0.059 [0.064] N=1,426	0.077 [0.067] N=1,247	0.029 [0.077] N=1,046
Mathematics	3.848	0.045 [0.080] N=1,802	0.079 [0.084] N=1,665	0.098 [0.090] N=1,498	0.132 [0.099] N=1,308	0.106 [0.109] N=1,100
Science	3.847	0.005 [0.084] N=1,294	-0.005 [0.089] N=1,188	0.042 [0.096] N=1,060	0.043 [0.103] N=927	0.015 [0.116] N=778
Foreign language	3.644	0.051 [0.088] N=1,460	0.102 [0.091] N=1,348	0.094 [0.099] N=1,214	0.095 [0.108] N=1,064	0.082 [0.122] N=892
Culture arts and sports	3.997	**0.133* [0.068] N=1,548**	**0.138* [0.071] N=1,428**	**0.132* [0.074] N=1,292**	0.129 [0.081] N=1,129	0.107 [0.092] N=948
Vocational and technological	3.840	0.096 [0.149] N=593	0.099 [0.157] N=542	0.116 [0.174] N=494	0.062 [0.186] N=434	0.012 [0.210] N=359

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.055 [0.089] N=1,539	-0.031 [0.095] N=1,400	0.025 [0.108] N=1,232	0.041 [0.126] N=1,039	-0.107 [0.140] N=824
Social studies and life sciences	0.032 [0.083] N=1,445	0.031 [0.085] N=1,314	0.055 [0.097] N=1,155	0.089 [0.110] N=976	-0.029 [0.111] N=775
Mathematics	0.049 [0.094] N=1,520	0.103 [0.105] N=1,383	0.142 [0.117] N=1,216	0.223 [0.137] N=1,026	0.190 [0.165] N=818
Science	-0.066 [0.099] N=1,098	-0.101 [0.109] N=992	-0.042 [0.122] N=864	-0.072 [0.139] N=731	-0.142 [0.163] N=582
Foreign language	0.000 [0.112] N=1,237	0.084 [0.121] N=1,125	0.051 [0.138] N=991	0.058 [0.156] N=841	0.025 [0.187] N=669
Culture arts and sports	0.115 [0.103] N=1,308	0.110 [0.113] N=1,188	0.111 [0.128] N=1,052	0.108 [0.158] N=889	0.036 [0.198] N=708
Vocational and technological	0.143 [0.190] N=515	0.153 [0.208] N=464	0.176 [0.255] N=416	0.146 [0.283] N=356	0.118 [0.344] N=281

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.920	0.057 [0.061] N=2,364	0.061 [0.066] N=2,157	0.058 [0.072] N=1,910	0.060 [0.078] N=1,640	0.054 [0.087] N=1,324
Social studies and life sciences	3.940	0.091 [0.060] N=2,245	0.086 [0.065] N=2,046	0.086 [0.071] N=1,816	0.071 [0.077] N=1,555	0.091 [0.087] N=1,256
Mathematics	3.870	-0.013 [0.056] N=2,338	0.007 [0.058] N=2,134	-0.001 [0.062] N=1,892	-0.013 [0.066] N=1,625	0.035 [0.075] N=1,313
Science	3.884	0.030 [0.066] N=1,735	0.036 [0.069] N=1,593	0.032 [0.074] N=1,410	0.035 [0.077] N=1,213	0.063 [0.082] N=986
Foreign language	3.672	0.056 [0.070] N=1,942	0.040 [0.074] N=1,781	0.008 [0.079] N=1,581	0.020 [0.084] N=1,354	0.012 [0.088] N=1,098
Culture arts and sports	4.006	0.029 [0.050] N=2,002	0.027 [0.052] N=1,833	0.051 [0.057] N=1,628	0.040 [0.059] N=1,391	0.072 [0.070] N=1,135
Vocational and technological	3.853	0.041 [0.118] N=790	0.113 [0.126] N=719	0.168 [0.140] N=626	0.228 [0.155] N=540	0.243 [0.173] N=446

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.066 [0.077] N=2,013	0.074 [0.087] N=1,806	0.074 [0.098] N=1,559	0.094 [0.105] N=1,289	0.120 [0.132] N=973
Social studies and life sciences	0.103 [0.082] N=1,914	0.095 [0.093] N=1,715	0.100 [0.107] N=1,485	0.086 [0.122] N=1,224	0.155 [0.154] N=925
Mathematics	-0.072 [0.081] N=1,988	-0.055 [0.086] N=1,784	-0.077 [0.096] N=1,542	-0.135 [0.104] N=1,275	-0.050 [0.130] N=963
Science	-0.020 [0.095] N=1,472	-0.034 [0.102] N=1,330	-0.048 [0.118] N=1,147	-0.071 [0.126] N=950	-0.041 [0.155] N=723
Foreign language	-0.006 [0.102] N=1,642	-0.053 [0.112] N=1,481	-0.131 [0.128] N=1,281	-0.165 [0.149] N=1,054	-0.268 [0.177] N=798
Culture arts and sports	-0.027 [0.068] N=1,690	-0.054 [0.072] N=1,521	-0.016 [0.085] N=1,316	-0.062 [0.086] N=1,079	-0.019 [0.123] N=823
Vocational and technological	-0.113 [0.143] N=676	-0.012 [0.156] N=605	0.065 [0.192] N=512	0.145 [0.201] N=426	0.190 [0.243] N=332

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.920	0.055 [0.060] N=2,364	0.060 [0.065] N=2,157	0.059 [0.070] N=1,910	0.065 [0.076] N=1,640	0.058 [0.086] N=1,324
Social studies and life sciences	3.940	0.090 [0.059] N=2,245	0.086 [0.065] N=2,046	0.089 [0.070] N=1,816	0.077 [0.076] N=1,555	0.096 [0.086] N=1,256
Mathematics	3.870	-0.013 [0.056] N=2,338	0.007 [0.058] N=2,134	-0.001 [0.062] N=1,892	-0.015 [0.066] N=1,625	0.033 [0.075] N=1,313
Science	3.884	0.033 [0.066] N=1,735	0.039 [0.069] N=1,593	0.036 [0.074] N=1,410	0.042 [0.076] N=1,213	0.070 [0.082] N=986
Foreign language	3.672	0.056 [0.069] N=1,942	0.042 [0.073] N=1,781	0.011 [0.076] N=1,581	0.031 [0.081] N=1,354	0.020 [0.085] N=1,098
Culture arts and sports	4.006	0.028 [0.051] N=2,002	0.026 [0.053] N=1,833	0.051 [0.057] N=1,628	0.042 [0.059] N=1,391	0.074 [0.071] N=1,135
Vocational and technological	3.853	0.032 [0.120] N=790	0.104 [0.127] N=719	0.162 [0.141] N=626	0.220 [0.157] N=540	0.235 [0.174] N=446

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.049 [0.076] N=2,013	0.057 [0.086] N=1,806	0.055 [0.096] N=1,559	0.075 [0.105] N=1,289	0.097 [0.131] N=973
Social studies and life sciences	0.090 [0.082] N=1,914	0.084 [0.093] N=1,715	0.089 [0.106] N=1,485	0.077 [0.121] N=1,224	0.141 [0.152] N=925
Mathematics	-0.068 [0.081] N=1,988	-0.051 [0.086] N=1,784	-0.074 [0.096] N=1,542	-0.132 [0.105] N=1,275	-0.046 [0.130] N=963
Science	-0.024 [0.095] N=1,472	-0.036 [0.102] N=1,330	-0.054 [0.117] N=1,147	-0.072 [0.126] N=950	-0.043 [0.155] N=723
Foreign language	-0.016 [0.100] N=1,642	-0.062 [0.109] N=1,481	-0.139 [0.124] N=1,281	-0.167 [0.145] N=1,054	-0.277 [0.175] N=798
Culture arts and sports	-0.037 [0.069] N=1,690	-0.064 [0.072] N=1,521	-0.026 [0.084] N=1,316	-0.072 [0.086] N=1,079	-0.033 [0.122] N=823
Vocational and technological	-0.131 [0.144] N=676	-0.029 [0.157] N=605	0.050 [0.192] N=512	0.119 [0.203] N=426	0.168 [0.248] N=332

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.920	0.053 [0.058] N=2,364	0.056 [0.063] N=2,157	0.054 [0.068] N=1,910	0.060 [0.072] N=1,640	0.055 [0.083] N=1,324
Social studies and life sciences	3.940	0.091 [0.055] N=2,245	0.086 [0.061] N=2,046	0.086 [0.065] N=1,816	0.077 [0.069] N=1,555	0.096 [0.080] N=1,256
Mathematics	3.870	-0.013 [0.055] N=2,338	0.005 [0.057] N=2,134	-0.006 [0.060] N=1,892	-0.019 [0.064] N=1,625	0.029 [0.074] N=1,313
Science	3.884	0.034 [0.064] N=1,735	0.042 [0.066] N=1,593	0.039 [0.071] N=1,410	0.047 [0.072] N=1,213	0.073 [0.079] N=986
Foreign language	3.672	0.052 [0.067] N=1,942	0.040 [0.072] N=1,781	0.007 [0.074] N=1,581	0.029 [0.079] N=1,354	0.022 [0.084] N=1,098
Culture arts and sports	4.006	0.034 [0.048] N=2,002	0.032 [0.050] N=1,833	0.056 [0.053] N=1,628	0.050 [0.055] N=1,391	0.087 [0.066] N=1,135
Vocational and technological	3.853	0.044 [0.115] N=790	0.116 [0.122] N=719	0.187 [0.134] N=626	**0.263* [0.147] N=540**	**0.312* [0.158] N=446**

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.049 [0.075] N=2,013	0.055 [0.086] N=1,806	0.053 [0.095] N=1,559	0.075 [0.102] N=1,289	0.103 [0.128] N=973
Social studies and life sciences	0.093 [0.078] N=1,914	0.086 [0.090] N=1,715	0.088 [0.100] N=1,485	0.081 [0.113] N=1,224	0.148 [0.143] N=925
Mathematics	-0.067 [0.079] N=1,988	-0.050 [0.085] N=1,784	-0.078 [0.093] N=1,542	-0.135 [0.101] N=1,275	-0.046 [0.127] N=963
Science	-0.027 [0.092] N=1,472	-0.036 [0.100] N=1,330	-0.050 [0.115] N=1,147	-0.063 [0.122] N=950	-0.031 [0.153] N=723
Foreign language	-0.018 [0.097] N=1,642	-0.058 [0.107] N=1,481	-0.138 [0.119] N=1,281	-0.163 [0.143] N=1,054	-0.267 [0.172] N=798
Culture arts and sports	-0.037 [0.069] N=1,690	-0.063 [0.072] N=1,521	-0.030 [0.084] N=1,316	-0.074 [0.088] N=1,079	-0.025 [0.123] N=823
Vocational and technological	-0.147 [0.139] N=676	-0.050 [0.151] N=605	0.051 [0.186] N=512	0.114 [0.204] N=426	0.232 [0.235] N=332

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.920	0.045 [0.058] N=2,364	0.046 [0.063] N=2,157	0.040 [0.067] N=1,910	0.046 [0.072] N=1,640	0.029 [0.083] N=1,324
Social studies and life sciences	3.940	0.083 [0.055] N=2,245	0.076 [0.061] N=2,046	0.074 [0.065] N=1,816	0.064 [0.069] N=1,555	0.073 [0.080] N=1,256
Mathematics	3.870	-0.022 [0.056] N=2,338	-0.007 [0.058] N=2,134	-0.018 [0.062] N=1,892	-0.029 [0.065] N=1,625	0.014 [0.076] N=1,313
Science	3.884	0.023 [0.063] N=1,735	0.026 [0.066] N=1,593	0.022 [0.071] N=1,410	0.030 [0.071] N=1,213	0.041 [0.078] N=986
Foreign language	3.672	0.036 [0.068] N=1,942	0.022 [0.073] N=1,781	-0.010 [0.075] N=1,581	0.015 [0.081] N=1,354	-0.004 [0.086] N=1,098
Culture arts and sports	4.006	0.025 [0.047] N=2,002	0.020 [0.049] N=1,833	0.040 [0.052] N=1,628	0.042 [0.054] N=1,391	0.070 [0.065] N=1,135
Vocational and technological	3.853	0.022 [0.110] N=790	0.093 [0.116] N=719	0.166 [0.127] N=626	**0.238* [0.141] N=540**	**0.262* [0.158] N=446**

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.054 [0.075] N=2,013	0.060 [0.085] N=1,806	0.057 [0.095] N=1,559	0.083 [0.102] N=1,289	0.087 [0.129] N=973
Social studies and life sciences	0.096 [0.078] N=1,914	0.089 [0.089] N=1,715	0.093 [0.100] N=1,485	0.088 [0.113] N=1,224	0.132 [0.144] N=925
Mathematics	-0.066 [0.081] N=1,988	-0.052 [0.087] N=1,784	-0.078 [0.096] N=1,542	-0.132 [0.105] N=1,275	-0.053 [0.136] N=963
Science	-0.025 [0.092] N=1,472	-0.037 [0.100] N=1,330	-0.053 [0.115] N=1,147	-0.057 [0.122] N=950	-0.048 [0.155] N=723
Foreign language	-0.027 [0.098] N=1,642	-0.067 [0.109] N=1,481	-0.142 [0.121] N=1,281	-0.158 [0.143] N=1,054	-0.286 [0.174] N=798
Culture arts and sports	-0.035 [0.067] N=1,690	-0.065 [0.070] N=1,521	-0.034 [0.081] N=1,316	-0.069 [0.086] N=1,079	-0.041 [0.124] N=823
Vocational and technological	-0.151 [0.133] N=676	-0.055 [0.146] N=605	0.065 [0.179] N=512	0.148 [0.191] N=426	0.221 [0.235] N=332

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.910	0.060 [0.062] N=2,228	0.071 [0.067] N=2,031	0.071 [0.071] N=1,795	0.078 [0.077] N=1,539	0.066 [0.088] N=1,247
Social studies and life sciences	3.928	**0.108* [0.060] N=2,119**	0.108 [0.065] N=1,930	0.117 [0.071] N=1,711	0.113 [0.079] N=1,463	0.113 [0.091] N=1,187
Mathematics	3.857	-0.007 [0.057] N=2,207	0.014 [0.058] N=2,013	0.029 [0.064] N=1,782	0.023 [0.070] N=1,528	0.059 [0.082] N=1,240
Science	3.867	0.036 [0.068] N=1,629	0.034 [0.070] N=1,496	0.040 [0.076] N=1,323	0.054 [0.081] N=1,137	0.052 [0.090] N=929
Foreign language	3.650	0.063 [0.070] N=1,832	0.043 [0.074] N=1,680	0.021 [0.077] N=1,489	0.049 [0.081] N=1,275	0.023 [0.088] N=1,040
Culture arts and sports	4.008	0.058 [0.053] N=1,917	0.051 [0.054] N=1,755	0.075 [0.059] N=1,555	0.067 [0.061] N=1,327	0.097 [0.074] N=1,088
Vocational and technological	3.852	0.090 [0.120] N=759	0.157 [0.129] N=692	0.221 [0.142] N=601	**0.259* [0.156] N=520**	**0.310* [0.175] N=432**

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.048 [0.078] N=1,894	0.065 [0.088] N=1,697	0.073 [0.096] N=1,461	0.104 [0.105] N=1,205	0.108 [0.134] N=913
Social studies and life sciences	0.108 [0.083] N=1,802	0.105 [0.094] N=1,613	0.127 [0.107] N=1,394	0.144 [0.121] N=1,146	0.162 [0.159] N=870
Mathematics	-0.074 [0.081] N=1,872	-0.057 [0.084] N=1,678	-0.040 [0.096] N=1,447	-0.069 [0.107] N=1,193	-0.011 [0.134] N=905
Science	-0.018 [0.097] N=1,377	-0.047 [0.101] N=1,244	-0.052 [0.113] N=1,071	-0.040 [0.123] N=885	-0.110 [0.156] N=677
Foreign language	-0.000 [0.103] N=1,543	-0.054 [0.112] N=1,391	-0.112 [0.125] N=1,200	-0.097 [0.144] N=986	-0.234 [0.178] N=751
Culture arts and sports	0.009 [0.075] N=1,615	-0.025 [0.077] N=1,453	0.017 [0.089] N=1,253	-0.023 [0.091] N=1,025	0.012 [0.127] N=786
Vocational and technological	-0.058 [0.149] N=647	0.039 [0.161] N=580	0.125 [0.195] N=489	0.164 [0.204] N=408	0.270 [0.256] N=320

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.910	0.055 [0.062] N=2,228	0.066 [0.067] N=2,031	0.067 [0.072] N=1,795	0.069 [0.078] N=1,539	0.061 [0.088] N=1,247
Social studies and life sciences	3.928	**0.104* [0.060] N=2,119**	0.104 [0.065] N=1,930	0.113 [0.072] N=1,711	0.104 [0.079] N=1,463	0.106 [0.089] N=1,187
Mathematics	3.857	-0.013 [0.057] N=2,207	0.009 [0.058] N=2,013	0.022 [0.064] N=1,782	0.012 [0.069] N=1,528	0.052 [0.080] N=1,240
Science	3.867	0.033 [0.068] N=1,629	0.031 [0.070] N=1,496	0.035 [0.076] N=1,323	0.044 [0.080] N=1,137	0.043 [0.087] N=929
Foreign language	3.650	0.061 [0.070] N=1,832	0.042 [0.073] N=1,680	0.019 [0.076] N=1,489	0.046 [0.080] N=1,275	0.023 [0.085] N=1,040
Culture arts and sports	4.008	0.054 [0.053] N=1,917	0.045 [0.054] N=1,755	0.068 [0.059] N=1,555	0.053 [0.061] N=1,327	0.078 [0.073] N=1,088
Vocational and technological	3.852	0.085 [0.123] N=759	0.148 [0.131] N=692	0.195 [0.144] N=601	0.231 [0.158] N=520	0.264 [0.176] N=432

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.041 [0.079] N=1,894	0.058 [0.089] N=1,697	0.064 [0.097] N=1,461	0.081 [0.107] N=1,205	0.097 [0.135] N=913
Social studies and life sciences	0.102 [0.082] N=1,802	0.101 [0.094] N=1,613	0.122 [0.107] N=1,394	0.128 [0.122] N=1,146	0.158 [0.156] N=870
Mathematics	-0.085 [0.081] N=1,872	-0.067 [0.084] N=1,678	-0.051 [0.096] N=1,447	-0.094 [0.106] N=1,193	-0.021 [0.133] N=905
Science	-0.022 [0.097] N=1,377	-0.049 [0.101] N=1,244	-0.052 [0.115] N=1,071	-0.044 [0.124] N=885	-0.097 [0.154] N=677
Foreign language	-0.003 [0.104] N=1,543	-0.056 [0.113] N=1,391	-0.113 [0.127] N=1,200	-0.104 [0.145] N=986	-0.225 [0.179] N=751
Culture arts and sports	0.008 [0.075] N=1,615	-0.026 [0.078] N=1,453	0.015 [0.090] N=1,253	-0.032 [0.093] N=1,025	0.003 [0.129] N=786
Vocational and technological	-0.051 [0.149] N=647	0.046 [0.161] N=580	0.120 [0.195] N=489	0.168 [0.204] N=408	0.262 [0.253] N=320

Notes: Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.